Examples of using WMS to access Simena

Introduction

The scope of this document is to determine the method of access to the information generated by Simena.

If you want to access a listing of available data, you should make a GetCapabilities request.

If you want to query the corrected temperature for a concrete observatory in Peru, for example, you should make a GetFeature request with the following values:

- `QUERY_LAYERS=MTG_CORRECTED_TEMPERATURE` (as taken from the GetCapabilities)
- Since the WMS protocol uses BBOX+WIDTH+HEIGHT+I+J, use `WIDTH=101&HEIGHT=101&I=5&J=50` with a small BBOX to query the desired coordinate. For example:
- Observatory coordinates are x=1.25343 y=40.435325
- BBOX centered on that point would be: x-0.000001,y+0.000001,x+0.000001,y-0.000001 -> `BBOX=1.25342,40.435326,1.25344,40.435324`

If you want to get a map, you should make a GetMap request. The meaning of the colours can be obtained by a GetLegend request.

WMS GetCapabilities example

Asking for the general information about the server available data. Request:
https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetCapabilities
Response (an WMS Capabilities XML document):


```xml
▼<WMS_Capabilities xmlns="http://www.opengis.net/wms" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instanc
 ▶<Service>
   ...
  </Service>
 ▼<Capability>
   ▶<Request>
     ...
    </Request>
   ▼<Exception>
      <Format>XML</Format>
      <Format>INIMAGE</Format>
      <Format>BLANK</Format>
    </Exception>
   ▼<Layer>
      <Name>meteo</Name>
      <Title>Weather forecast</Title>
      <Abstract/>
      <CRS>EPSG:4326</CRS>
      <CRS>EPSG:3857</CRS>
      <CRS>EPSG:25830</CRS>
     ▼<EX_GeographicBoundingBox>
        <westBoundLongitude>-180</westBoundLongitude>
        <eastBoundLongitude>180</eastBoundLongitude>
        <southBoundLatitude>-90</southBoundLatitude>
        <northBoundLatitude>90</northBoundLatitude>
      </EX_GeographicBoundingBox>
      <BoundingBox CRS="CRS:84" minx="-178.04821777413082" miny="-87.15158463433839" maxx="177.6145019524
      <BoundingBox CRS="EPSG:4326" minx="-178.04821777413082" miny="-87.15158463433839" maxx="177.614501
      <BoundingBox CRS="EPSG:25830" minx="-3.6120028414793307E8" miny="-2.837919762899621E8" maxx="7.718
      <BoundingBox CRS="XEPSG:4326" minx="-87.15158463433839" miny="-178.04821777413082" maxx="87.661926
     ▼<Layer queryable="0" opaque="0" cascaded="0">
        <Name>METEOGRID_REGIONS</Name>
        <Title>Regional boundaries</Title>
        <Abstract>None</Abstract>
       ▼<Style>
          <Name>default</Name>
          <Title>default</Title>
         ▼<LegendURL width="233" height="625">
            <Format>image/png</Format>
            <OnlineResource xmlns:xlink="http://www.w3.org/1999/xlink" xlink:type="simple" xlink:href="htt
          </LegendURL>
        </Style>
      </Layer>
     ▼<Layer queryable="1" opaque="0" cascaded="0">
        <Name>HARMONIE_ARPEGE_ECMWF_TMAX</Name>
        <Title>Temperature</Title>
        <Abstract><b>Origen de la información</b>: Modelos Harmonie-Arpege-ECMWF. Pasada de las 0:00 UTC<
        <b>Resolución temporal</b>: Diaria<br> <b>Horizonte temporal</b>: 8 días</Abstract>
       ▼<Style>
          <Name>default</Name>
          <Title>default</Title>
         ▼<LegendURL width="233" height="625">
            <Format>image/png</Format>
            <OnlineResource xmlns:xlink="http://www.w3.org/1999/xlink" xlink:type="simple" xlink:href="htt
```


WMS GetFeature example

Asking for the evolution of a variable in a certain coordinate. As per WMS standard, the requested point is relative (coordinates I, J of a WIDTH x HEIGHT) to a bounding box. A TOLERANCE parameter is added in order to enclose near point features. Request:
https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetFeatureInfo&QUERY_LAYERS=ARPEGE_TEMPERATURE&LAYERS=ARPEGE_TEMPERATURE&STYLES=&TIME=2023-01-19T12%3A00%3A00Z&INFO_FORMAT=application%2Fjson&TOLERANCE=3&HORIZON=__all__&I=405&J=60&WIDTH=512&HEIGHT=512&CRS=EPSG%3A3857&BBOX=-1252344.271424327%2C3757032.814272983%2C6.984919309616089e-10%2C5009377.085697311
Legible values:

SERVICE: WMS

VERSION: 1.3.0

REQUEST: GetFeatureInfo

QUERY_LAYERS: ARPEGE_TEMPERATURE

LAYERS: ARPEGE_TEMPERATURE

STYLES:

TIME: 2023-01-19T12:00:00Z

INFO_FORMAT: application/json

TOLERANCE: 3

HORIZON: __all__

I: 405

J: 60

WIDTH: 512

HEIGHT: 512

CRS: EPSG:3857

BBOX: -1252344.271424327,3757032.814272983,6.984919309616089e-10,5009377.085697311

In this case, all temporal horizons from the calculation run in which said TIME is contained are retrieved.

Response (a GeoJSON document):


```
{
   "type": "FeatureCollection",
   "features": [
       {
           "type": "FeatureCollection",
           "features": [
               {
                   "type": "Feature",
                   "name": "ARPEGE_TEMPERATURE",
                   "geometry": null,
                   "properties": {
                       "base_time": "2023-01-19T00:00:00Z",
                       "value": -0.04648923873901367,
                       "HOR_ARPEGE": 0,
                       "time": "2023-01-19T00:00:00Z"
                   }
               },
               {
                   "type": "Feature",
                   "name": "ARPEGE_TEMPERATURE",
                   "geometry": null,
                   "properties": {
                       "base_time": "2023-01-19T00:00:00Z",
                       "value": -0.24009323120117188,
```

```
                        "HOR_ARPEGE": 1,
                        "time": "2023-01-19T01:00:00Z"
                    }
                },
…
            ],
            "layer_name": "ARPEGE_TEMPERATURE",
            "layer_title": "ARPEGE_TEMPERATURE",
            "has_time": true
        }
    ]
}
```


When reading point features (as opposed to raster) the poi can be specified by id with the poi_id optional parameter.

WMS GetMap example

Asking for the coloured map of a certain Bounding Box. Request:
https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetMap&FORMAT=image%2Fpng&TRANSPARENT=true&LAYERS=ARPEGE_TEMPERATURE&STYLES=&TIME=2023-01-19T12%3A00%3A00Z&WIDTH=512&HEIGHT=512&CRS=EPSG%3A3857&BBOX=-1252344.271424327%2C3757032.814272983%2C6.984919309616089e-10%2C5009377.085697311

Legible values:

SERVICE: WMS

VERSION: 1.3.0

REQUEST: GetMap

FORMAT: image/png

TRANSPARENT: true

LAYERS: ARPEGE_TEMPERATURE

STYLES:

TIME: 2023-01-19T12:00:00Z

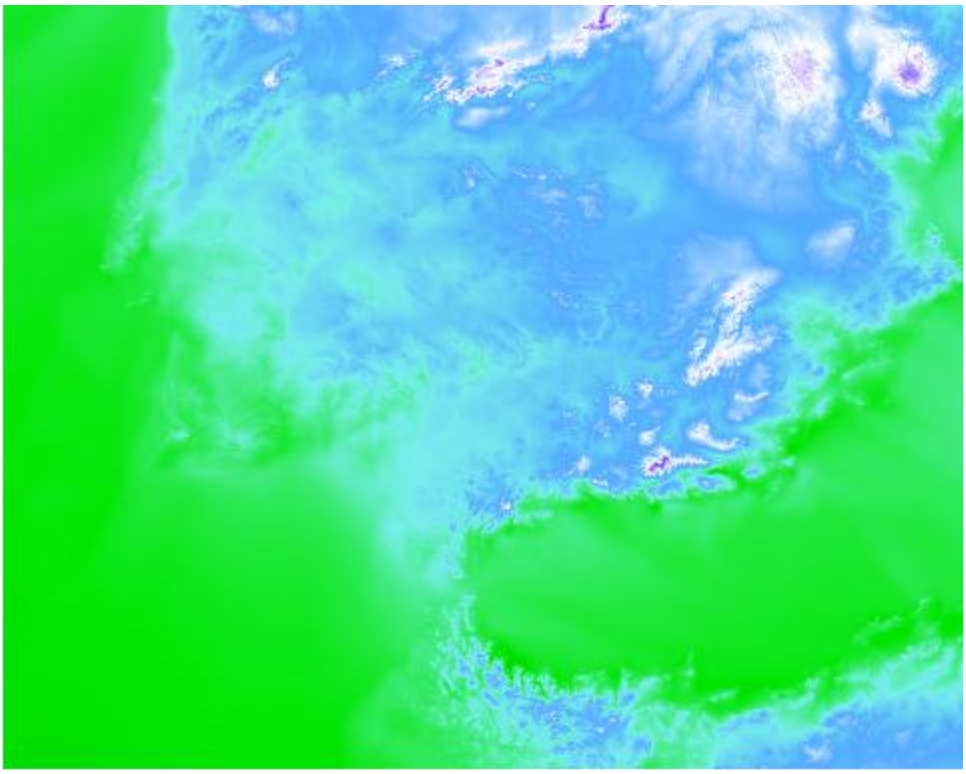
WIDTH: 512

HEIGHT: 512

CRS: EPSG:3857

BBOX: -1252344.271424327,3757032.814272983,6.984919309616089e-10,5009377.085697311

Response (an image):



WMS GetLegend example

Asking for the legend of said map. Request:

https://agcommander.simena.red/wms/wms/?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetLegendGraphic&FORMAT=image%2Fpng&LAYER=ARPEGE_TEMPERATURE&width=150&TRANSPARENT=TRUE

SERVICE: WMS

VERSION: 1.3.0

REQUEST: GetLegendGraphic

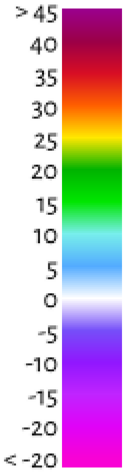
FORMAT: image/png

LAYER: ARPEGE_TEMPERATURE

width: 150

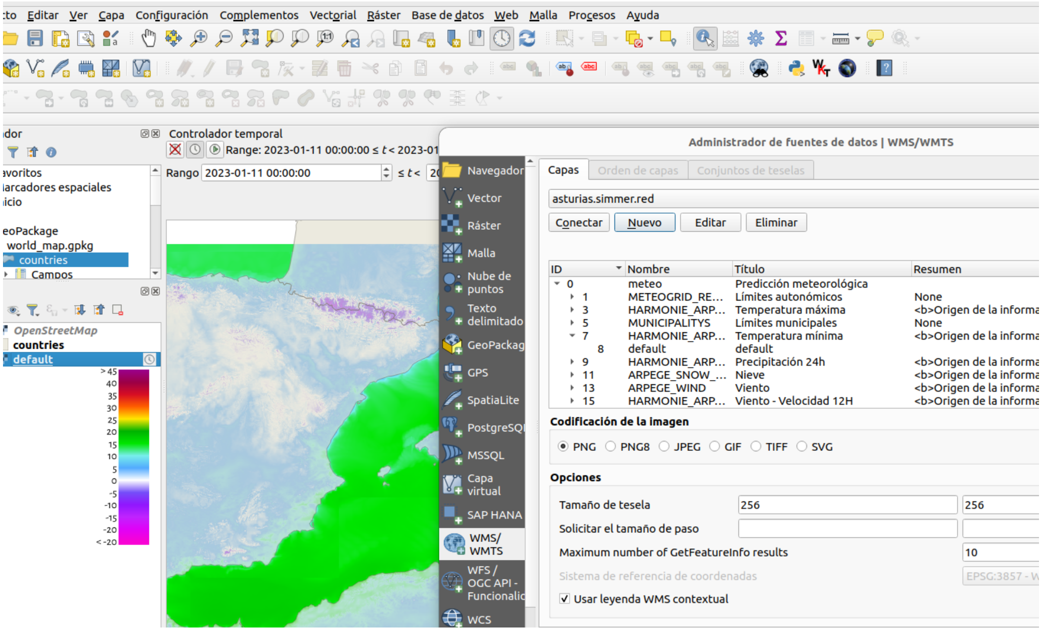
TRANSPARENT: TRUE

Response (an image):



Use in Desktop Client

The WMS Service can be used from desktop clients such as QGIS, ESRI, etc.



References

https://www.ogc.org/standards/wms
https://en.wikipedia.org/wiki/Web_Map_Service